

MIS 443 – Finance Analysis with PostgreSQL

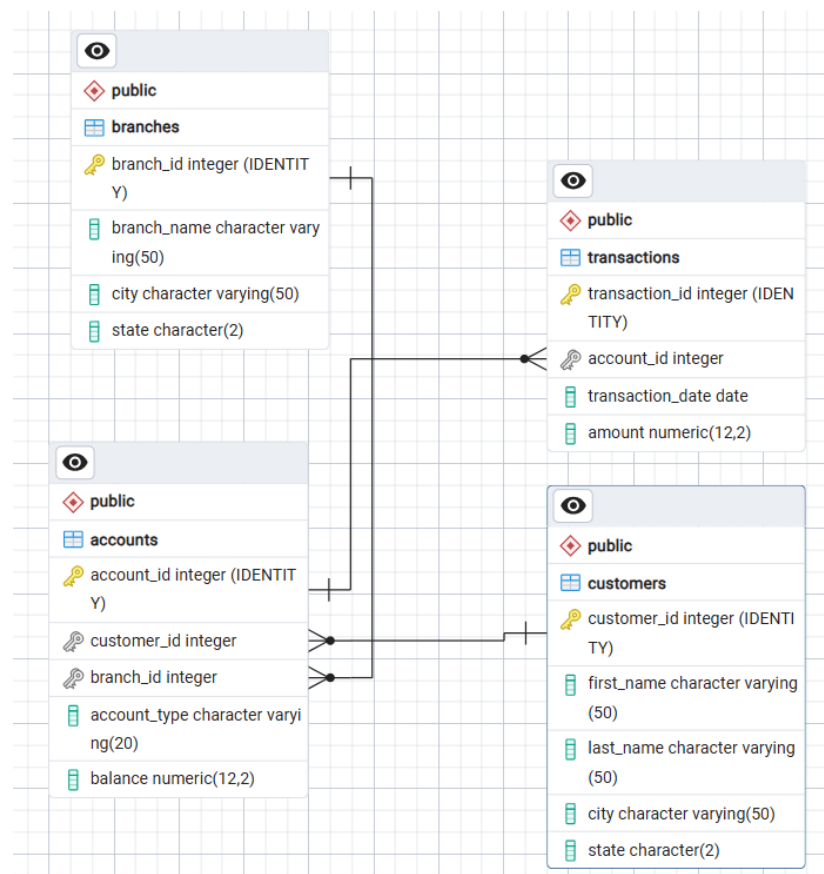
Student Information

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Question 1 – Database Setup

A PostgreSQL database was created using the student's full name in lowercase without spaces or Vietnamese diacritics. The provided SQL script (**MIS443_Finance_PostgreSQL.sql**) was executed successfully, and all required tables were created in the public schema.

Display ERD



The database consists of:

- **Customers:** 6 records
- **Branches:** 15 records
- **Accounts:** 15 records
- **Transactions:** 15 records

The ERD shows that each customer can own multiple accounts, each branch manages multiple accounts, and each account can contain multiple transactions.

Question 2 – Customer and Account Overview

The query identified two customers living in New York: John Doe and Jane Doe. The bank currently manages 15 accounts, providing an overview of the institution's customer portfolio.

Question 3 – Account Balance Analysis

The total balance of all Checking accounts is 31,000.00. For customers living in Los Angeles, Michael Lee has the highest combined balance (60,000.00), followed by Jennifer Wang with 15,000.00, indicating that Michael Lee has the strongest financial portfolio in that region.

Question 4 – Branch and Customer Portfolio Analysis

The North Beach branch in San Francisco has the highest average account balance of 30,000.00, demonstrating strong account performance. The customer owning the single highest-balance account is Michael Lee, whose Savings account holds 50,000.00.

Question 5 – Customer Value and Activity

The most active customers are Jane Doe and Alice Johnson, each with 4 transactions, indicating the highest level of banking activity. Excluding Credit Card accounts, Michael Lee has the highest combined Checking and Savings balance of 60,000.00, making him the bank's most valuable deposit customer.

Question 6 – Advanced Finance Analysis

The North Beach branch records the highest total balance across all account types, with 60,000.00. Customer ranking based on total balance places Michael Lee first (60,000.00), followed by Alice Johnson (25,000.00) and Bob Smith (20,500.00). Using a Common Table Expression (CTE), the analysis also shows that the Main and South Bay branches have the highest transaction activity, each recording 4 transactions.

Conclusion

This analysis demonstrates how PostgreSQL can be used to manage and analyze banking data through SQL queries involving filtering, join, aggregation, subqueries, window functions, and Common Table Expressions (CTEs). The results provide valuable insights into customer value, account balances, branch performance, and transaction activity, supporting more effective financial and operational decision-making.

Github link: [https://github.com/tuyetnguyenthianhbbs23-collab/MIS-443/tree/main/MIS%20443 Mock%20Test](https://github.com/tuyetnguyenthianhbbs23-collab/MIS-443/tree/main/MIS%20443%20Mock%20Test)